# Course Title: Technology, Environment and Society

## HUM 4147

# Content

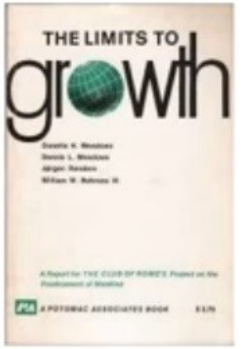
- Sustainability and Sustainable Development
- Environmental Stewardship
- Environmental Justice and Equity

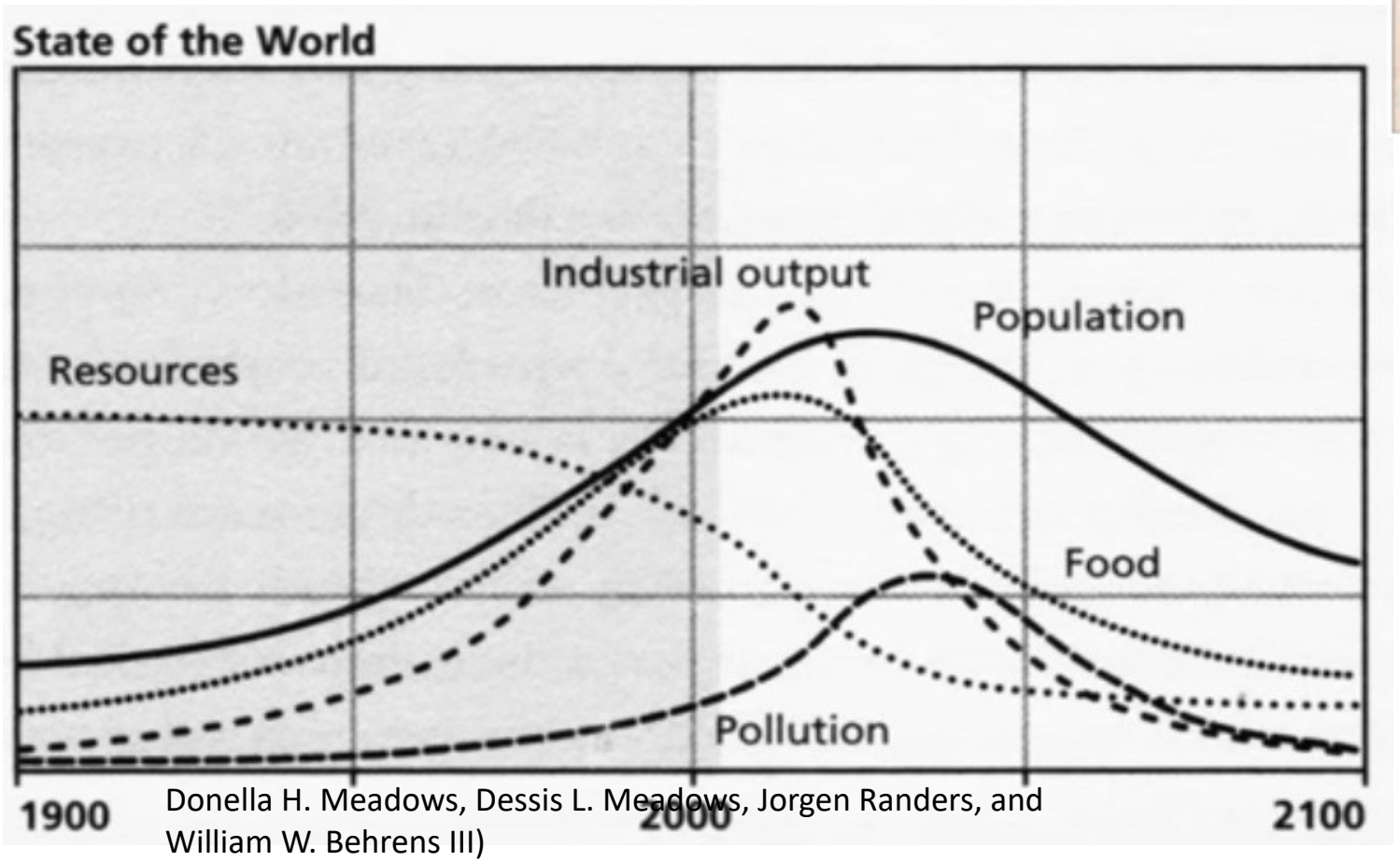
# Sustainability and Sustainable Development



# **Sustainability** and Sustainable Development

Need for sustainable development  came after team of scientist from MIT, meet in  1972




Donella H. Meadows, Dessis L. Meadows, Jorgen Randers, and William W. Behrens III)

# Sustainability and **Sustainable Development**

"Sustainable development meets the needs of the present without compromising the ability of future generations to meet their own need"

The World Commission on Environment and Development, United Nations, 1987

Arises many questions…..what will we need?

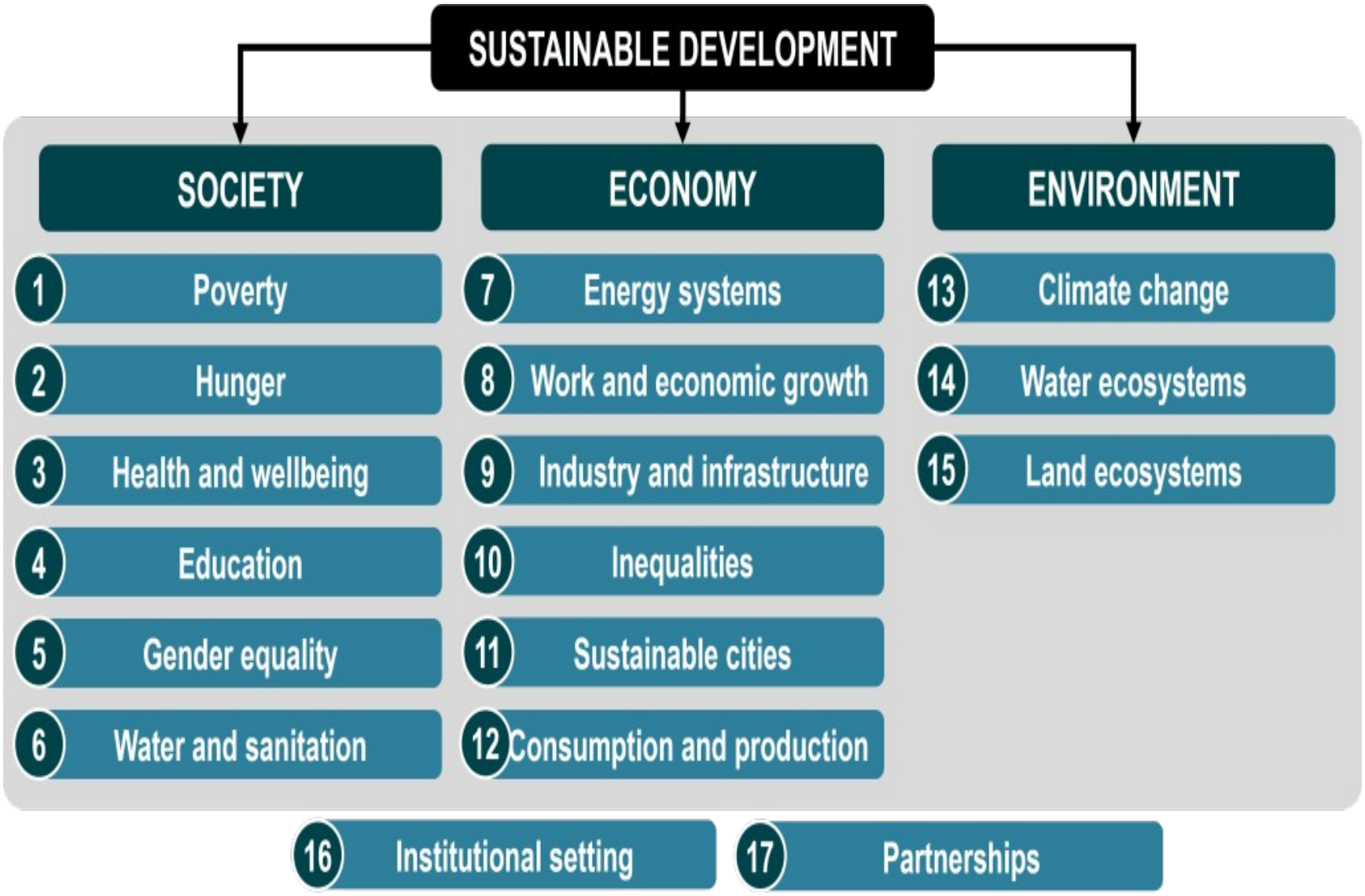
We know that we are using resources very fast

## The United Nations' Sustainable Development Goals

- In 2015, the United Nations drew up 17 Sustainable Development Goals to set the course for the future of our world. With the 2030 Agenda, 193 countries have committed to ensuring that all people can live in dignity by 2030 while at the same time preserving the natural foundations of life in the long term, whether social, economic, or ecological.

# The United Nations' Sustainable Development Goals




SUSTAINABLE DEVELOPMENT
SOCIETY
ECONOMY
ENVIRONMENT
1 Poverty
2 Hunger
3 Health and wellbeing
4 Education
5 Gender equality
6 Water and sanitation
7 Energy systems
8 Work and economic growth
9 Industry and infrastructure
10 Inequalities
11 Sustainable cities
12 Consumption and production
13 Climate change
14 Water ecosystems
15 Land ecosystems
16 Institutional setting
17 Partnerships

- Each of these goals can be associated with societal, economic, or environmental improvements:
- **Society**. These goals prioritize satisfying the diversified needs of the population, such as food, health, and education being among the most basic. Maintaining human capital (knowledge, skills, and capabilities) is mostly the responsibility of educational systems, but corporations also provide substantial training opportunities to their workforce.
- **Economy**. These goals promote improvements in the welfare of populations. Key concepts are related to achieving or sustaining economic growth, maximizing profits, increasing competitiveness, and expanding markets. Globalization has given a new dimension to economic development by enabling an extended range of comparative advantages. However, like all economic processes, globalization promotes growth differently as regions and social classes capture their opportunities differently. This has led to inequalities.
- **Environment.** These goals concern the footprint of human activities on environmental systems, notably their carrying capacity. The overarching issue of climate change remains salient, particularly over the issue of carbon emissions.

# Sustainability and Sustainable Development

**Sustainability**

Stocks and Flow → Exp.; expenses ≤ income

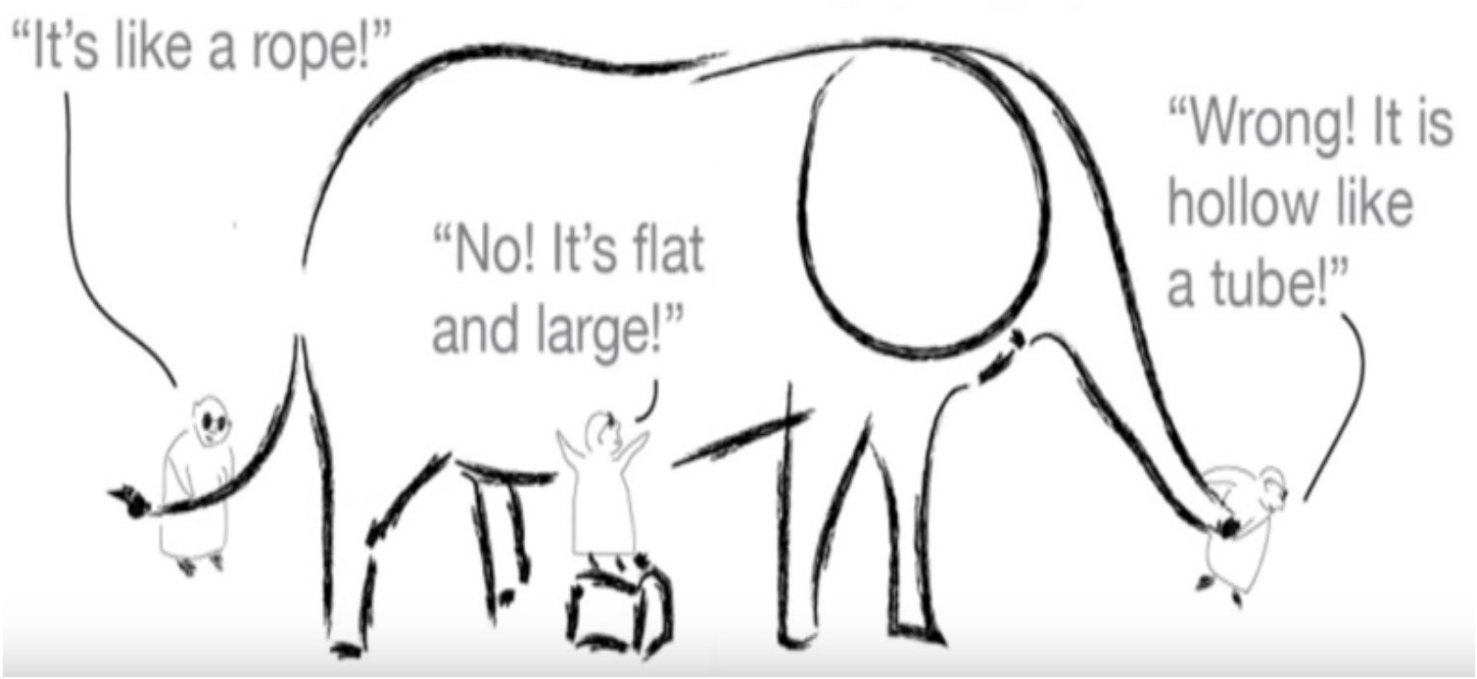
**Development**

Well-being → Can mean different things to different people

Comfort, happiness, health

# Sustainability and Sustainable Development

## The Parable of the Elephant and 3 Blind Men




Multiple perspectives are required to understand

# **Sustainability** and Sustainable Development

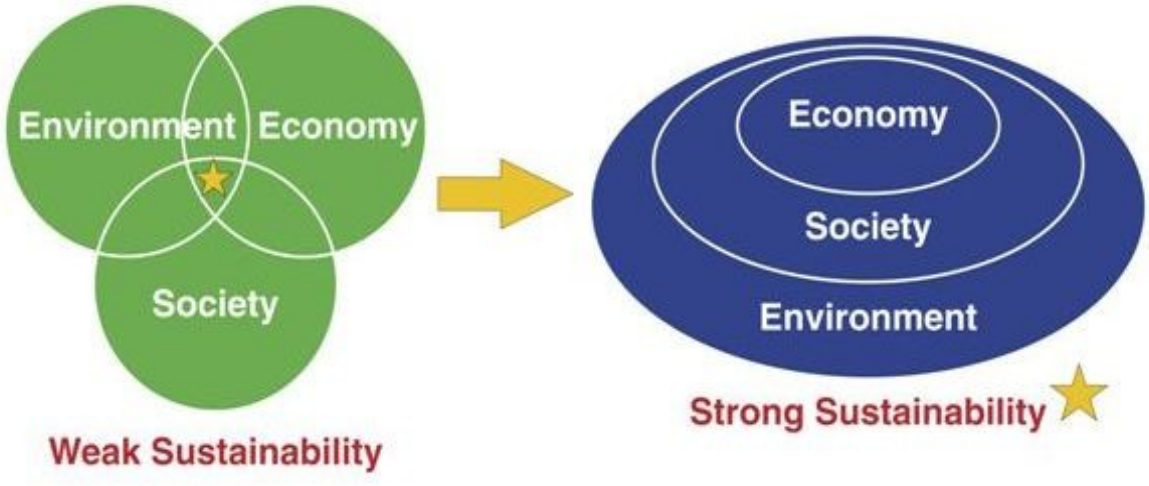
Often time sustainability viewed as a balance between Economy, Environment and Society – **traditional model**

**False Notion**

- Economic activity outside the environment
- inherent competition
- separate goals

**Alternative Model**

- One environment
- society develops inside the env.
- Economic activity




Source: Maureen Hart - Sustainable Measures

# 5 Important Measures for Sustainable Development

## (i) Technology

Using appropriate technology is one which is locally adaptable, eco-friendly, resource efficient and culturally suitable.



This concept is known as "design with nature"

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# Sustainability and Sustainable Development

**(ii) Reduce, Reuse, and Recycle Approach**

The 3-R approach advocating minimization of resource use, using them again and again instead of passing it on to the waste stream

**(iii) Promoting Environmental Education and Awareness**

Making environmental education the centre of all learning process will greatly help in changing the attitude of people towards the earth and the environment.

feeling of belongingness to earth

'Earth thinking'

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# Sustainability and Sustainable Development

**(iv) Resource Utilization as Per Carrying Capacity:**

If the carrying capacity of a system is crossed (say, by over exploitation of a resource), environmental degradation starts and continues till it reaches a point of no return.

**Carrying capacity has two basic components:**

i. Supporting capacity i.e. the capacity to regenerate

ii. Assimilative capacity i.e. the capacity to tolerate different stresses.

**(v) Improving Quality of Life Including Social, Cultural and Economic Dimensions:**

Ensuring the distribution of fare share of benefits

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# Environmental Stewardship



# Environmental Stewardship

**Environmental Stewardship**

Stewardship of the environment refers to **protecting the environment** through **recycling, conservation, regeneration, and restoration**.

It means **taking responsibility for our choices**.
The responsibility for environmental quality should be shared by all whose actions affect the environment.

# Environmental Stewardship



# Environmental Stewardship

Individual or Organizations practice:-

✓ Use resources efficiently

✓ Reduce waste and minimize the negative impacts on the environment

✓ Translate reduced consumption and waste into financial savings

✓ Provide a healthy, more efficient and effective working environment

# Environmental Stewardship

- ✓ Embodies the concept of a Green Office, a Green School or Green Home which is conceptualized as smarter and better as it is:
- ❑ Ecological - using non-toxic, recycled, environmentally friendly products and supplies
- ❑ Efficient – using as little energy and other resources as possible and putting out the smallest amount of waste as possible
- ❑ Healthy – will generate as little visual, noise and physical pollution as possible

# Principles of Environmental Stewardship

**Principle 1: Resource Conservation** – making the most efficient use of resources (materials, energy and water) through:

- Energy Efficiency and Conservation
- Water Efficiency and Conservation
- Solid Waste Management and Conceptualizing Waste as a Resource

**Principle 2: Pollution Prevention** – minimizing contamination of the environment by chemicals or other materials through:

- Vehicle/Fleet Management
- Management of the Use of Chemicals and Other Hazardous Substances

**Principle 3: Occupational Health and Safety**

- Managing Indoor Air Quality
- Comfort and Productivity in the Workplace
- Emergency and Disaster Management

# Environmental Stewardship

## Benefits of Implementing an Environmental Stewardship

- ✓Economic Benefits
  - ✓Health Benefits
    - ✓Protection and Conservation of the Environment



# Environmental Justice and Equity



# Environmental Justice and Equity

.. is the **fair treatment** and **meaningful involvement** of all people regardless of race, color, national origin, or income with respect to the development, implementation, and enforcement of environmental laws, regulations, and policies.

# Environmental Justice and Equity

**Fair treatment means**

….that no group of people should bear a disproportionate share of the negative environmental consequences resulting from industrial, governmental and commercial operations or policies.

# Environmental Justice and Equity

## Meaningful involvement means

- people have an opportunity to participate in decisions about activities that may affect their environment and/or health;
- the public contribution can influence the regulatory agency's decision;
- their concerns will be considered in the decision making process;
- the decision makers seek out and facilitate the involvement of those potentially affected.

# Environmental Justice and Equity

## What is the Problem?

- The impact of environmental policies and decisions have differentiated impacts across population groups

- Most countries have utterly no idea what environmental justice is nor have any mechanism for addressing EJ issues

**Examples**

- Haiti and much of the developed Western Hemisphere
- Environmental neglect in New Orleans
- The former East German Republic and its neighbors
- Philippines Islands and Japan
- Sub-Saharan Africa and Europe

# Environmental Justice and Equity

**National Policy Alternatives**

- Improving information exchange. Knowledge is power and leverage
- National or state environmental justice legislation or regulation
- Non-governmental action, groups or agenda

**Outcomes and Futures**

- EJ resolution cannot solve all the world's social/economic justice problems
- International EJ issues may be the next great world policy issue